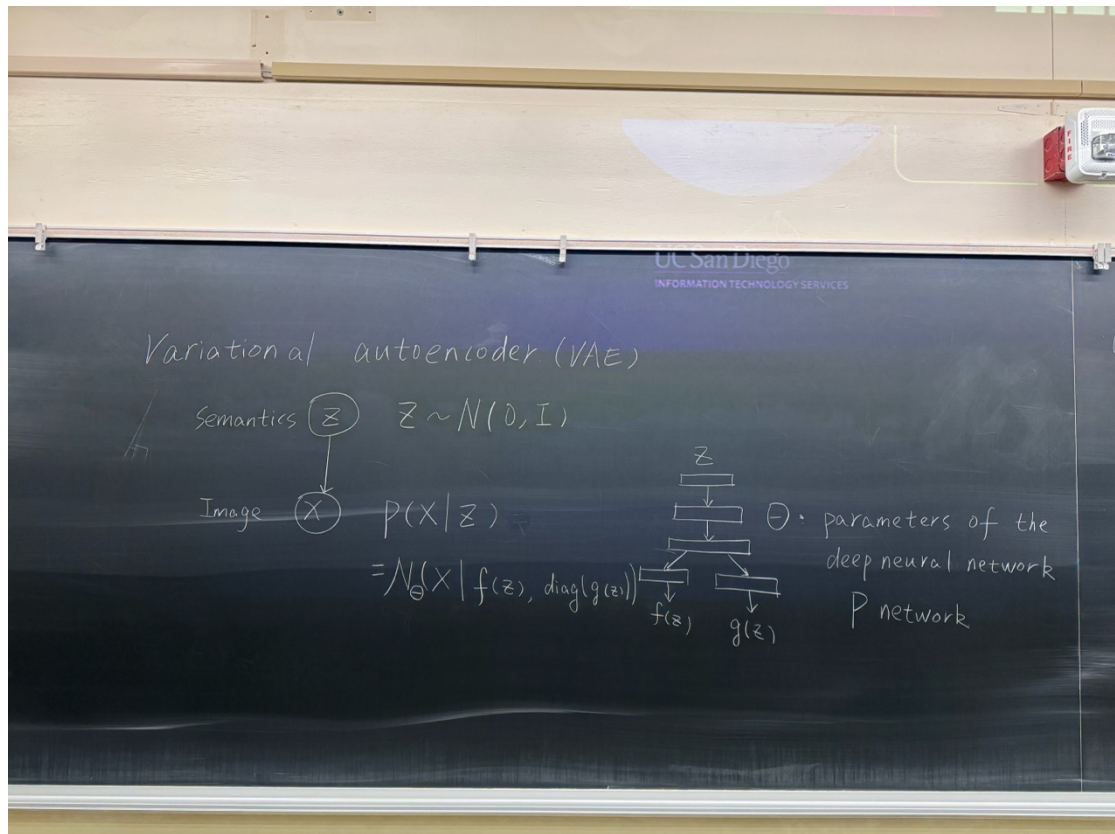




① Variational inference

$p(z|x)$ data understanding

$q_{\lambda}(z)$

$p(z|x) \approx q_{\lambda}(z)$

$\min_{\lambda} KL(q_{\lambda}(z) || p(z|x))$

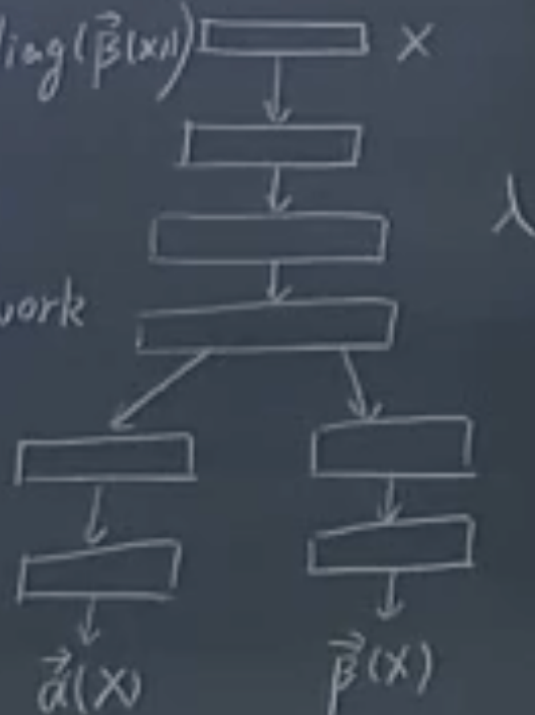
$= \min_{\lambda} -E_{q_{\lambda}(z)} [\log \frac{p(z,x)}{p(x|z)p(z)}] + E_{q_{\lambda}(z)} [\log q_{\lambda}(z)]$

$$q(z) = N(z | \vec{\mu}, \Sigma)$$

$$q(z|x)$$

$$= N(z | \vec{\alpha}(x), \text{diag}(\vec{\beta}(x)))$$

Q network



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$$-E_{q_{\lambda}(z|x)}[\log p_{\theta}(z, x)] + E_{q_{\lambda}(z|x)}[\log q_{\lambda}(z|x)]$$

$p_{\theta}(x|z)$ P network
 θ
 $p_{\theta}(z, x) = p_{\theta}(x|z) p(z)$

Q network λ
 $q_{\lambda}(z|x)$
 P network θ
 $p_{\theta}(x|z)$

Define
Par

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network θ
 $p(z)$

Define the model
 ↓
 Parameter learning (training)
 ↓
 Use the trained model

Generate data
 Inference

Moment matching X
 Maximum likelihood ✓
 $X_1, X_2, \dots, X_N$ Data likelihood
 $p(x)$
 $= \int_z p(x, z) dz$
 $= \int_z p_{\theta}(x|z) p(z) dz$
 θ^*

$\max_{\theta} \prod_{n=1}^N p_{\theta}(x_n)$

Moment matching X

Maximum likelihood ✓

$X_1, X_2, \dots, X_N$

Data likelihood

$$p(X)$$
$$= \int_{\mathbf{z}} p(X, \mathbf{z}) d\mathbf{z}$$

$$= \int_{\mathbf{z}} p_{\theta}(X|\mathbf{z}) p(\mathbf{z}) d\mathbf{z}$$

$$\max_{\theta} \prod_{n=1}^N p_{\theta}(X_n)$$

$$\log \prod_{n=1}^N p_{\theta}(X_n)$$

$$= \sum_{n=1}^N \log p_{\theta}(X_n)$$

$$\sum_{n=1}^N \log p_{\theta}(X_n)$$

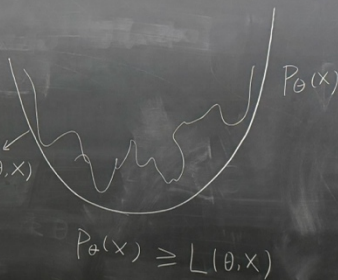
$$\log p_{\theta}(X) = \log \int_{\mathbf{z}} p_{\theta}(X, \mathbf{z}) d\mathbf{z}$$

$$= \log \int_{\mathbf{z}} \left[\frac{p_{\theta}(X, \mathbf{z})}{q(\mathbf{z})} \right] q(\mathbf{z}) d\mathbf{z} \quad L(\theta, X)$$

$$= \log E_{q(\mathbf{z})} \left[\frac{p_{\theta}(X, \mathbf{z})}{q(\mathbf{z})} \right]$$

$$\geq E_{q(\mathbf{z})} \left[\log \frac{p_{\theta}(X, \mathbf{z})}{q(\mathbf{z})} \right]$$

$$= L(\theta, X)$$

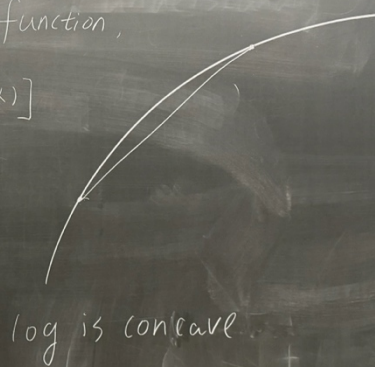
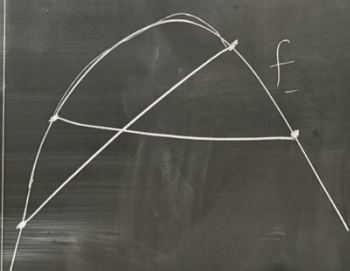


$$p_{\theta}(X) \geq L(\theta, X)$$

Jensen's inequality

If f is a concave function,

$$f(E[x]) \geq E[f(x)]$$



$\log$ is concave

$$\sum_{n=1}^N \log p_{\theta}(x_n)$$

$$\log p_{\theta}(x) = \log \int_{\mathbf{z}} p_{\theta}(x, \mathbf{z}) d\mathbf{z}$$

$$= \log \int_{\mathbf{z}} \frac{p_{\theta}(x, \mathbf{z})}{q(\mathbf{z})} q(\mathbf{z}) d\mathbf{z}$$

$$= \log E_{q(\mathbf{z})} \left[\frac{p_{\theta}(x, \mathbf{z})}{q(\mathbf{z})} \right]$$

$$\geq E_{q(\mathbf{z})} \left[\log \frac{p_{\theta}(x, \mathbf{z})}{q(\mathbf{z})} \right]$$

$$= \mathcal{L}(\theta, x)$$

$$= E_{q(\mathbf{z})} [\log p_{\theta}(x, \mathbf{z}) - \log q(\mathbf{z})]$$

$$= E_{q(\mathbf{z})} [\log p_{\theta}(x, \mathbf{z})] - E_{q(\mathbf{z})} [\log q(\mathbf{z})]$$

Evidence lower bound (ELBO)

$$\max_{\theta} \sum_{n=1}^N (E_{q(z)} [\log p_{\theta}(x_n, z)] - E_{q(z)} [\log q(z)])$$

Define q

the same as the q distribution in variational inference

$$\max_{\theta, \lambda} E_{q_{\lambda}(z|x)} [\log p_{\theta}(x, z)] - E_{q_{\lambda}(z|x)} [\log q_{\lambda}(z|x)]$$

$$\max_{\lambda, \theta} L(\lambda, \theta)$$

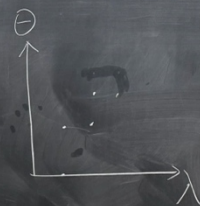
Coordinate ascent

while not converged

E step 1: Fix θ , optimize λ

M step 2: Fix λ , optimize θ

Expectation maximization



$$\max_{\lambda}$$

$$= \min_{\lambda}$$

$$\max_{\lambda} E_{q_{\lambda}(z|x)} [\log p_{\theta^*}(x, z)] - E_{q_{\lambda}(z|x)} [\log q_{\lambda}(z|x)]$$

$$= \min_{\lambda} - ($$



imization